four squares in the two middle columns. The area that belongs to all three literals is the single square in the first row and third column.

- In a similar manner, the other three squares belonging to the function F are marked by 1's in the map.
- The next step is to subdivide the given area into adjacent squares. These are indicated in the map by two rectangles, each enclosing two 1's. The upper right rectangle represents the area enclosed by $x'y$; the lower left, the area enclosed by xy'. The sum of these two terms gives the answer:

$$\boxed{F = x'y + xy'}$$

Example 3-2: Simplify the Boolean function:

$$F = x'yz + xy'z' + xyz + xyz'$$

$x \backslash yz$	00	01	11	10
0			1	
1	1		1	1

Fig 3.5: Map for Example 3-2

There are four squares marked with 1's, one for each minterm of the function. Two adjacent squares are combined in the third column to give a two-literal term yz.

- The remaining two squares with 1's are also adjacent by the new definition and are shown in the diagram enclosed by half rectangles. These two squares, when combined, give the two-literal term xz'. The simplified function becomes:

$$\boxed{F = yz + xz'}$$

Consider now any combination of four adjacent squares in the three-variable map. Any such combination represents the ORing of four adjacent minterms and results in an expression of only one literal.

- **Example:** The sum of the four adjacent minterms m_0, m_2, m_4 and m_6 reduces to the single literal z' as shown:

$$x'y'z' + x'yz' + xy'z' + xyz' = (x'y'z' + x'yz') + (xy'z' + xyz') \quad \text{(Theorem 4(a))}$$

$$= (z'\cdot(x'y') + z'\cdot(x'y)) + (z'\cdot(xy') + z'\cdot(xy)) \quad \text{(Postulate 3(a))}$$

$$= ((z'x')\cdot y' + (z'x')\cdot y) + ((z'x)\cdot y' + (z'x)\cdot y) \quad \text{(Theorem 4(b))}$$